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Communication-Avoiding Krylov Subspace Methods for Option Pricing

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in the School of Mathematics

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Project Code: <https://github.com/kevinknights29/caksm>



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Write acknowledgements to your supervisor, classmates, friends, family, partner... anyone who supported you during the MSc.

ABSTRACT

Write a summary of your project.

Keywords: keyword1, keyword2, keyword3,...

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2 INTRODUCTION

After spatial discretization, the semi-discrete system takes the affine form

$$\frac{d\mathbf{u}}{d\tau} = \mathbf{A}\mathbf{u} + \mathbf{b}, \quad \mathbf{u}(\tau_0) = \mathbf{u}_0. \quad (2.0.1)$$

Following Niesen & Wright, the inhomogeneity is absorbed into an *extended system*: define

$$\tilde{\mathbf{A}} = \begin{bmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{0}^T & 0 \end{bmatrix} \in \mathbb{R}^{(N+1) \times (N+1)}, \quad \tilde{\mathbf{v}} = \begin{bmatrix} \mathbf{u}_k \\ 1 \end{bmatrix} \in \mathbb{R}^{N+1},$$

so that the exact solution at time $\tau_0 + h$ is encoded in the first N components of $e^{h\tilde{\mathbf{A}}}\tilde{\mathbf{v}}$.

The Krylov approximation proceeds via the Arnoldi iteration on $\tilde{\mathbf{A}}$ with starting vector $\tilde{\mathbf{v}}$, producing an orthonormal basis $\mathbf{V}_m$ and an upper Hessenberg matrix $\bar{\mathbf{H}}_m \in \mathbb{R}^{(m+1) \times m}$. Dropping the last row gives $\mathbf{H}_m \in \mathbb{R}^{m \times m}$, and the approximation reads

$$e^{h\tilde{\mathbf{A}}}\tilde{\mathbf{v}} \approx \beta \mathbf{V}_m e^{h\mathbf{H}_m} \mathbf{e}_1, \quad \beta = \|\tilde{\mathbf{v}}\|. \quad (2.0.2)$$

Algorithm 1 provides the full adaptive procedure.

The dominant cost of the Krylov exponential integrator is the Arnoldi iteration (Algorithm 2). Each step j requires:

1. One sparse matrix–vector product $\mathbf{w} = \tilde{\mathbf{A}}\mathbf{v}_j$, costing $\mathcal{O}(N)$.
2. Orthogonalization against all previous vectors $\mathbf{v}_1, \dots, \mathbf{v}_j$, costing $\mathcal{O}(jN)$.

Crucially, step j cannot begin until $\mathbf{v}_j$ is available, which in turn requires step $j - 1$ to have completed. This creates a *strictly sequential data dependency* that prevents straightforward parallelization of the Arnoldi iteration across the Krylov dimension m .

While the individual SpMV and inner products within each step can be parallelized (data-parallel), this “inner parallelism” scales poorly beyond a moderate number of cores due to communication overhead and synchronization barriers in the orthogonalization phase.

We discuss three complementary strategies, each targeting a different aspect of the computation.

2.1 STRATEGY 1: COMMUNICATION-AVOIDING KRYLOV METHODS (S-STEP / CA-ARNOLDI)

The classical Arnoldi algorithm performs one SpMV and one global synchronization (for the inner products) per iteration. Communication-avoiding (CA) Krylov methods, also known as *s-step* methods, break this bottleneck by computing s basis vectors simultaneously before orthogonalizing them as a block.

The key idea is to replace the sequential computation $\mathbf{v}, \mathbf{A}\mathbf{v}, \mathbf{A}^2\mathbf{v}, \dots, \mathbf{A}^s\mathbf{v}$ (which is embarrassingly parallel in the SpMV sense) with a single communication phase. The resulting “basis” is then orthogonalized using a tall-skinny QR (TSQR) factorization, which requires only $\mathcal{O}(\log p)$ messages on p processors instead of the $\mathcal{O}(s)$ synchronizations of the classical method.

Algorithm 1: Adaptive Krylov Exponential Integrator for $\mathbf{u}' = \mathbf{A}\mathbf{u} + \mathbf{b}$

Input: Matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$, vector $\mathbf{b} \in \mathbb{R}^N$, initial condition $\mathbf{u}_0$, final time T , time-step $\Delta\tau$, tolerance tol

Output: Solution snapshots $\{\mathbf{u}_k\}_{k=0}^K$ at $\tau_k = k \Delta\tau$

```

1  $\tau \leftarrow 0$ ;
2 for  $k = 0, 1, \dots, K - 1$  do
3    $h \leftarrow \min(\Delta\tau, T - \tau)$ ;
4    $m \leftarrow 1$ ;  $\text{err} \leftarrow \infty$ ;

   /* Form extended system */
5    $\tilde{\mathbf{A}} \leftarrow \begin{bmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{0}^T & 0 \end{bmatrix} \in \mathbb{R}^{(N+1) \times (N+1)}$ ;
6    $\tilde{\mathbf{v}} \leftarrow \begin{bmatrix} \mathbf{u}_k \\ 1 \end{bmatrix} \in \mathbb{R}^{N+1}$ ;

   /* Adaptive Krylov loop */
7   while  $m \leq N + 1$  do
8     Run  $m$  steps of Arnoldi on  $(\tilde{\mathbf{A}}, \tilde{\mathbf{v}}) \Rightarrow \mathbf{V}_{m+1}, \bar{\mathbf{H}}_m$  // Algorithm 2
9     Store  $\bar{h}_{m+1,m} \leftarrow [\bar{\mathbf{H}}_m]_{m+1,m}$  // sub-diagonal entry
10     $\mathbf{H}_m \leftarrow \bar{\mathbf{H}}_m(1:m, 1:m)$  // drop last row
11     $\beta \leftarrow \|\tilde{\mathbf{v}}\|$ ;
12     $\mathbf{f} \leftarrow e^{h\mathbf{H}_m} \mathbf{e}_1$  // small  $m \times m$  matrix exponential
13     $\mathbf{w} \leftarrow \beta \mathbf{V}_m \mathbf{f}$  // Krylov approximation in full space

    /* A posteriori error estimate */
14     $\text{err} \leftarrow |\bar{h}_{m+1,m}| \cdot \|\mathbf{f}\|_m$ ;
15    if  $\text{err} < \text{tol}$  then
16      break;
17    end
18     $m \leftarrow m + 1$ ;
19  end
20   $\mathbf{u}_{k+1} \leftarrow \mathbf{w}(1:N)$  // extract first  $N$  components
21   $\tau \leftarrow \tau + h$ ;
22 end

```

Algorithm 2: Arnoldi Iteration

Input: Matrix $\tilde{\mathbf{A}} \in \mathbb{R}^{(N+1) \times (N+1)}$, vector $\tilde{\mathbf{v}} \in \mathbb{R}^{N+1}$, subspace dimension m , breakdown tolerance tol

Output: Orthonormal basis $\mathbf{V}_{m+1} \in \mathbb{R}^{(N+1) \times (m+1)}$, upper Hessenberg $\bar{\mathbf{H}}_m \in \mathbb{R}^{(m+1) \times m}$

```

1  $\beta \leftarrow \|\tilde{\mathbf{v}}\|$ ;  $\mathbf{v}_1 \leftarrow \tilde{\mathbf{v}}/\beta$ ;
2 for  $j = 1, \dots, m$  do
3    $\mathbf{w} \leftarrow \tilde{\mathbf{A}} \mathbf{v}_j$ ;
4   for  $i = 1, \dots, j$  do
5     // Modified Gram-Schmidt
6      $h_{i,j} \leftarrow \mathbf{v}_i^T \mathbf{w}$ ;
7      $\mathbf{w} \leftarrow \mathbf{w} - h_{i,j} \mathbf{v}_i$ ;
8   end
9    $h_{j+1,j} \leftarrow \|\mathbf{w}\|$ ;
10  if  $h_{j+1,j} < \text{tol}$  then
11    break // Invariant subspace found (happy breakdown)
12  end
13   $\mathbf{v}_{j+1} \leftarrow \mathbf{w}/h_{j+1,j}$ ;
14 end
```

Advantages for our problem:

- $\mathbf{A}$ is sparse with $\mathcal{O}(1)$ nonzeros per row, so the s SpMV's are cheap and highly parallelizable.
- The matrix is symmetric (or near-symmetric), so the Lanczos variant can be used, reducing storage and orthogonalization cost.
- The Krylov dimension m is typically small ($m \approx 10\text{--}30$), so even modest values of $s = 3\text{--}5$ can significantly reduce communication.

Challenges:

- The computed basis $\{\mathbf{v}, \mathbf{A}\mathbf{v}, \dots, \mathbf{A}^s \mathbf{v}\}$ becomes increasingly ill-conditioned as s grows (the vectors converge toward the dominant eigenvector). This limits s in practice.
- Newton or Chebyshev polynomial bases can mitigate the conditioning issue, at the cost of additional tuning parameters.

2.2 STRATEGY 2: RATIONAL KRYLOV SUBSPACES

Rather than using the polynomial Krylov subspace $\mathcal{K}_m(\mathbf{A}, \mathbf{v}) = \text{span}\{\mathbf{v}, \mathbf{A}\mathbf{v}, \dots, \mathbf{A}^{m-1}\mathbf{v}\}$, a *rational Krylov subspace* uses shifted inverses:

$$\mathcal{Q}_{m+1}(\mathbf{A}, \mathbf{v}, q_m) = q_m(\mathbf{A})^{-1} \text{span}\{\mathbf{v}, \mathbf{A}\mathbf{v}, \dots, \mathbf{A}^m \mathbf{v}\}, \quad (2.2.1)$$

where $q_m(z) = \prod_{j=1}^m (z - \xi_j)$ and the ξ_j are called *poles* or *shifts*. When the poles are distinct and not eigenvalues of $\mathbf{A}$, an equivalent characterization is

$$\mathcal{Q}_{m+1}(\mathbf{A}, \mathbf{v}) = \text{span}\{\mathbf{v}, (\mathbf{A} - \xi_1 I)^{-1} \mathbf{v}, (\mathbf{A} - \xi_2 I)^{-1} \mathbf{v}, \dots, (\mathbf{A} - \xi_m I)^{-1} \mathbf{v}\}.$$

The crucial observation is that the vectors $(\mathbf{A} - \xi_j I)^{-1} \mathbf{v}$ for different shifts ξ_j can be computed *independently and in parallel*. This replaces the sequential data dependency of the polynomial Arnoldi iteration with m independent linear solves.

Connection to exponential integration: Rational Krylov subspaces are known to provide superior approximations to $e^{h\mathbf{A}}\mathbf{v}$ compared to polynomial Krylov subspaces of the same dimension, particularly when the eigenvalues of $\mathbf{A}$ span several orders of magnitude—precisely the situation arising from PDE discretizations. For a matrix with eigenvalues in $(-\infty, 0]$, rational approximants to e^z (such as Padé approximants) converge geometrically, whereas polynomial approximants converge only algebraically.

Advantages for our problem:

- The shifted systems $(\mathbf{A} - \xi_j I)\mathbf{x} = \mathbf{v}$ share the same sparsity pattern as $\mathbf{A}$. If a direct solver is used, the symbolic factorization can be reused across shifts; only the numerical factorization must be repeated.
- The matrix $\mathbf{A}$ is symmetric negative semi-definite, so choosing real negative shifts $\xi_j < 0$ ensures that $(\mathbf{A} - \xi_j I)$ is symmetric negative definite, hence nonsingular. Iterative solvers (e.g. CG with a suitable preconditioner) are applicable.
- With p processors and m poles, the ideal parallel speedup is $\min(p, m)$, limited only by the orthogonalization phase.

Challenges:

- The naive parallel construction (computing all shifted solves from the same starting vector $\mathbf{v}$) yields a basis whose condition number can grow *exponentially* with m , leading to loss of orthogonality in finite arithmetic (Berljafa & Güttel, 2017).
- The *near-optimal continuation pairs* framework of Berljafa & Güttel addresses this by choosing continuation vectors that keep the pre-orthogonalized basis well-conditioned, at the cost of a cheap approximate solve per iteration.
- The choice and number of poles ξ_j requires problem-specific tuning. For the matrix exponential, asymptotically optimal poles on the negative real axis are known (related to best rational approximations to e^z on $(-\infty, 0]$).

2.3 STRATEGY 3: COMBINED APPROACH

The two strategies above are not mutually exclusive. One could apply s -step techniques *within* a rational Krylov framework:

1. Partition the m poles into groups of size p (the number of processors).
2. Within each group, solve the p shifted systems in parallel (rational Krylov parallelism).
3. Use s -step techniques to reduce communication in the orthogonalization phase across groups.

Table ?? summarizes the three strategies.

Table 1: Comparison of parallelization strategies for Krylov exponential integration.

Strategy	Parallelism	Communication	Stability
Classical Arnoldi	Inner (SpMV, dots)	$\mathcal{O}(m)$ syncs	Stable
s -step / CA-Arnoldi	Block SpMV	$\mathcal{O}(m/s)$ syncs	s -dependent
Rational Krylov	Across shifts	$\mathcal{O}(m/p)$ syncs	Requires care
Combined	Both	Minimal	Problem-specific

3 SECTION EXAMPLE

Here is how you would reference a section, such as Section 2. You can cite an equation so it appears as a hyperlink equation number in parentheses (i.e., brackets) via (??).

3.1 SUBSECTION EXAMPLE

Here is another equation,

$$By = c \tag{3.1.1}$$